

BRIAN TEMU

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EDUCATION

University of Maryland Baltimore County

Master's in Data Science

May 2025

Maryland

University of Dar es salaam

Bachelor of Science in Computer Science

Nov 2019 – Oct 2022

Dar es salaam

Technical Skills

Programming: Python, JavaScript (TypeScript), SQL

Machine Learning: Deep Learning, Transformers (LLMs), NLP, and, Retrieval-Augmented Generation (RAG)

Frameworks/Libraries: PyTorch, TensorFlow, JAX, MLX, Hugging Face Transformers

Cloud/DevOps: Azure, Google Cloud, Vercel, Docker, Git, and Linux

Web/Software Engineering: FastAPI, React (Next.js), PostgreSQL, REST APIs, VS Code

WORK EXPERIENCE

Institute of Genome Science, University of Maryland Baltimore

Machine Learning/Software Engineer Intern

June 2025 – Sept 2025

Maryland

- Developed and implemented forecasting algorithms to accurately predict future trends in **time series data** using statistical and **machine learning models**
- Collaborated with research and engineering teams** to prototype and deploy predictive analytics tools, enhancing data-driven decision-making within the institution.
- Designed and optimized a **scalable time series** data platform, improving data ingestion and processing speed for large-scale scientific datasets using influxDB.

Milestones, Inc

Conversational AI Engineer

Apr 2025 – Aug 2025

California

- Developed scalable** FastAPI microservice architecture with **PostgreSQL backend**, handling concurrent voice sessions, automated call recording, transcription workflows, and comprehensive analytics tracking.
- Architected and deployed a **hybrid conversational AI system** integrating OpenAI Realtime API, ElevenLabs TTS, and Twilio Voice for real-time phone-based customer interactions.
- Implemented **intelligent conversation flows** with dynamic system prompting, customer data integration, human agent transfer capabilities, and multi-modal feedback collection via SMS.

Institute of Genome Science, University of Maryland Baltimore

Data Science Intern

May 2024 – Aug 2024

Maryland

- Automated** large-scale **antiSMASH** analyses using **Python scripts** and **shell automation**, identifying biosynthetic patterns in Bacterial Vaginosis **gene clusters**.
- Leveraged **HPC/grid computing** to run parallelized bioinformatics workflows, accelerating genetic marker discovery for targeted Bacteria Vaginosis research.
- Collaborated with researchers**, aligning computational results with study objectives to refine Bacteria Vaginosis genetic investigations.

Softnet Technologies Ltd

Software Engineer

Apr 2022 – Aug 2023

Dar es salaam

- Developed and optimized new features using **React** and **TypeScript**, improving user engagement by **15%**.
- Debugged and refactored** front-end code, reducing production bugs by **25%**, enhancing app stability.
- Led workshops on **TailwindCSS** and Figma, achieving **90%** adoption, reducing reliance on third-party templates.
- Implemented **Agile/Scrum** workflow, achieving **20%** more on-time project deliveries with **95%** sprint goal success.

- **Engineered** high-quality image datasets through data collection, preprocessing, and augmentation using **Python**, **OpenCV**, and **PIL**, ensuring optimal deep learning model performance.
- Developed and fine-tuned deep learning models with **PyTorch**, leveraging **CNN architectures** to improve feature extraction and classification.
- **Applied transfer learning** with **pretrained models** (e.g., **ResNet**, **EfficientNet**), increasing classification accuracy by **12%**.

PROJECTS

Eda | Your AI Learning Assistant

May 2025

- Developed an **AI-powered** learning assistant that transforms static documents into interactive study sessions, enabling users to query in a **vector database** and receive instant answers from uploaded PDFs and text files.
- Engineered a streamlined document processing pipeline integrating **NLP** techniques to extract and analyze content, enhancing the platform's responsiveness and accuracy.

CineRecall | Movie Identification Platform

Feb 2025

- Developed a dynamic movie recommendation platform using **Next.js** for the frontend and **Flask** for the backend, enabling seamless user interaction.
- Implemented **NLP** techniques to match user inputs with relevant films, enhancing **recommendation** accuracy.

Vision Transformer | Paper Replication

December 2023

- Reproduced and implemented key components of **Transformer architecture and attention mechanisms** in **PyTorch**, ensuring modularity and scalability.
- **Developed custom PyTorch modules** for multi-head self-attention, positional encoding, and feedforward layers, aligning with research paper specifications.

Baltimore Police Department Crime | Data Analysis

November 2023

- Analyzed crime trends using **Pandas** and **NumPy**, identifying patterns and correlations in crime rate fluctuations over time.
- **Visualized** key insights with **Matplotlib** and **Seaborn**, creating heatmaps, time-series plots, and bar charts to highlight trends and anomalies.

Real-Time Face-mask Detection System | Computer Vision

July 2022

- Developed a **YOLOv5-based object detection system**, achieving **97%** accuracy on a curated dataset through extensive **model fine-tuning and augmentation**.
- **Built an interactive UI with Streamlit**, enabling real-time visualization and user interaction with detected objects across different scenes.

Awards & Scholarships

Clark-Winchcole Foundation Scholarship

September 2024

Awarded by the Universities at Shady Grove in recognition of academic excellence within a USG program.

CERTIFICATION

DeepLearning.AI TensorFlow Developer Professional Certificate, Coursera

March 2023